

Research Methodology

Course Title: Research Methodology

Course No: RSM354

Nature of the Course: Theory

Semester: VI

Full Marks: 60+40

Pass Marks: 24+16

Credit Hrs: 3

Course Description:

This course introduces the concepts of research methodology. It also emphasizes on formulating research questions, conducting literature review, data collection, data analysis based on research questions and other various ethical issues, plagiarism.

Course Objectives:

This course focuses on the concept of research methodology; research design, formulation of research questions, process of review of literature, methods of data collection, measurement issues, ethical issues, plagiarism, data analysis and interpretations. After completion of the course, students will also be able to prepare a mini research report following standard notions of research methodology.

Course Contents:

Unit 1: Introduction to Research Methodology (4 Hrs.)

Meaning, objectives, motivation of research, concept of deductive and inductive theory, characteristics of scientific method, understanding of concept, construct, variable; research process

Unit 2: Research Design (6 Hrs.)

Concept of research design and its importance, features of a good research design, exploratory research design, descriptive research design, experimental research design, quantitative & qualitative research, empirical research, case study, applications of different research designs and their limitations, concept of dependent variable, independent variable and assessment of their relationships, issue of generalization of research findings

Unit 3: Identification and Formulation of Research Question (5 Hrs.)

Identification of research questions and its formulation, research objectives, statement of the problem, setting up the research hypothesis, sources of hypothesis and its utilities

Unit 4: Review of Literature (10 Hrs.)

Concept of review of literature, process of reviewing books, journal articles, reports; citation and referencing, an overview of different formats of citation and referencing, citation of books, reports, journal articles in American Psychological Association(APA) format, preparation of a small report based on review work on some relevant specific topic, use of encyclopedias, research guides, handbook, academic databases for computer Science and information technology field, use of reference management software such as Zotero/Mendeley, paper formatting such as LaTeX/MS Office, iThenticate Software for detection of Plagiarism

Unit 5: Methods of Data Collection and Sampling Techniques (7 Hrs.)

Primary and secondary data, method of collecting primary data, preparation of questionnaire, type of questions, characteristics of good questionnaire, concept of sampling, population, sampling frame, sampling and non-sampling error, probability and non-probability sampling, brief overview of simple random sampling, stratified sampling, cluster sampling, systematic sampling, multistage sampling and their practical applications in research problems, sample size estimation for estimating mean and proportion

Unit 6: Measurement and Scaling (3 Hrs.)

Concept of measurement, different levels of measurement, measurement of variables in likert scales, the issue of validity and reliability in research

Unit 7: Data Analysis and Report preparation (10 Hrs.)

Presentation of data through frequency tables, bar charts, pie charts, percentages, cross tabulations, summary measures, use of inferential statistical analysis in bivariate analysis(t-test, chi-square test, correlations, etc.) and concept of multivariate analysis, format of academic research report, preparation of a mini-research report following standard research methodology; ethical issues related to use of data, plagiarism, informed consent, self-privacy while collecting primary data, etc.

References:

1. Bryman A. & Bell E.(2015). *Business Research Methods*. Oxford University Press
2. Day R. A. & Gastel B. (2016). *How to write and publish a scientific paper*. (8th ed.). Greenwood Publication.
3. Kerlinger F.N. (2004). *Foundations of Behavioral Research*. Surgeet Publications, New Delhi, India
4. Kothari C.R. (2004). *Research Methodology*. New Age International (P) Limited, Publishers, India
5. Kumar, Ranjit (2011). *Research Methodology*. SAGE Publications Pvt. Limited, India.